

# Electric Vehicle Technologies and Applications Report

## Modelling and Simulation of a Battery-Electric Vehicle: Speed and Acceleration Characteristics, Driving Profile and Range

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### Abstract

This report presents a longitudinal model of a commercially available battery-electric vehicle, the Tesla Model 3 RWD (Highland, 2024), implemented in Python. The model resolves the complete energy chain from the road surface to the battery terminals: road load, tyre adhesion including dynamic axle-load transfer, the traction-machine torque and power envelope, a four-term drive-unit loss model, and a battery represented by a state-of-charge dependent open-circuit voltage in series with an internal resistance. Speed and acceleration characteristic curves are derived from the force balance and plotted. The WLTC Class 3b driving profile is implemented and the vehicle is simulated over it at 1 Hz, yielding a cycle consumption of 11.45 kWh/100 km and a predicted range on a full battery of 508 km, against a published WLTP figure of 513 km. Standing-start acceleration to 100 km/h is predicted at 6.14 s against a published 6.1 s, and the top speed is reproduced exactly. All three published figures are matched from a single parameter set with no per-result tuning. Of the 49 parameters the model requires, 32 are not published by the manufacturer and had to be estimated; every one of them is marked **in red** throughout this report, and a sensitivity study quantifies how much each changes the result. Three findings emerge: the vehicle is traction-limited rather than torque-limited off the line; its top speed is set by software rather than by physics; and the advertised drag coefficient understates the road load that actually governs range by 29%.

## 1 Introduction

The range of a battery-electric vehicle is not a property of its battery alone. It is the outcome of a chain in which every link loses something: the tyres and the air resist motion, the drive unit converts electrical power into shaft torque imperfectly, the battery dissipates part of its own output as heat, and the cabin draws power whether the vehicle is moving or not. A model that is useful for prediction has to represent all of them, because the weakest assumption in the chain sets the accuracy of the answer [1].

This report follows the semester project brief for *Electric Vehicle Technologies and Applications*. The brief asks the author to select a commercially available electric vehicle, derive and plot its speed and acceleration characteristic curves, choose and implement an appropriate driving profile, calculate the range available from a full battery on that profile, and make logical assumptions for the constants that are not published, marking them in red.

The vehicle selected is the Tesla Model 3 RWD (Highland, 2024 model year). It was chosen for three reasons. It is the best-selling electric vehicle of its class, so it is representative rather than exotic. It is certified under the Worldwide

Harmonised Light Vehicles Test Procedure (WLTP), so a published range figure exists against which a prediction can be tested. And it has been measured independently often enough that the parameters the manufacturer withholds can be estimated from more than guesswork.

The central methodological difficulty is that Tesla publishes neither a motor efficiency map, nor the gear ratio, nor the usable battery capacity. Thirty-two of the forty-nine parameters the model requires are therefore engineering estimates. A result resting on that many assumptions is only defensible if two things are done. First, every assumption must be stated with its basis, which is the purpose of the red marking the brief requires. Second, the influence of each assumption on the final answer must be quantified, so that the reader can see whether the conclusion rests on the values that were guessed with the least confidence. Section 9 does exactly that, and the answer is reassuring: the parameters that were hardest to pin down turn out to matter least.

The model is validated against the three figures the manufacturer publishes and cannot easily misstate, namely the standing-start acceleration time, the top speed, and the certified WLTP range. None of the three was used to calibrate

the model. Agreement on all three from one parameter set is the evidence offered that the model is sound.

## 1.1 Modelling Approaches

Vehicle propulsion models divide broadly into forward-facing and backward-facing formulations [1]. A forward-facing model places a driver controller in the loop, which demands a throttle input and propagates it through the powertrain to a resulting speed; it is the appropriate choice when the control strategy itself is the object of study. A backward-facing, quasi-static model instead takes the speed trace as given and asks what the powertrain must have done to produce it. For energy and range work the backward-facing form is standard, because the speed trace is prescribed by the legislative cycle and no driver model is needed to reproduce it. That formulation is adopted here.

The second methodological choice concerns the granularity of the loss description. Much of the introductory literature represents the drive unit by a single constant efficiency. That is adequate for a coarse estimate but misleading over a legislative cycle, because such cycles spend the majority of their time at light load, where a real machine is markedly less efficient than at its rated point [2]. This report therefore uses a four-term physical loss model, calibrated to representative operating points, in preference to a constant.

The third choice concerns the road-load description, and is treated at length in Section 3, because the difference between the textbook decomposition and the coastdown measurement used for type approval proves to be larger than several of the effects the model is careful to represent.

## 1.2 Nomenclature

Table 1 lists the symbols used throughout.

Table 1: Nomenclature.

| Sym.            | Unit                                              | Quantity                                |
|-----------------|---------------------------------------------------|-----------------------------------------|
| $m$             | kg                                                | Vehicle test mass                       |
| $\lambda$       | –                                                 | Rotational mass factor                  |
| $g$             | $\text{m/s}^2$                                    | Standard gravity                        |
| $v$             | $\text{m/s}$                                      | Vehicle speed                           |
| $a$             | $\text{m/s}^2$                                    | Vehicle acceleration                    |
| $\alpha$        | rad                                               | Road gradient angle                     |
| $f_r$           | –                                                 | Rolling resistance coefficient          |
| $\rho$          | $\text{kg/m}^3$                                   | Air density                             |
| $C_d$           | –                                                 | Aerodynamic drag coefficient            |
| $A$             | $\text{m}^2$                                      | Frontal area                            |
| $F_0, F_1, F_2$ | $\text{N}, \text{N s/m}, \text{N s}^2/\text{m}^2$ | Coastdown coefficients                  |
| $\mu$           | –                                                 | Peak tyre friction coefficient          |
| $\chi$          | –                                                 | Rear static load fraction               |
| $h$             | m                                                 | Centre-of-gravity height                |
| $L$             | m                                                 | Wheelbase                               |
| $N_r$           | N                                                 | Rear axle normal load                   |
| $T$             | N m                                               | Motor shaft torque                      |
| $\omega$        | rad/s                                             | Motor angular speed                     |
| $\omega_b$      | rad/s                                             | Base (corner-point) speed               |
| $i_g$           | –                                                 | Gear ratio                              |
| $\eta_g$        | –                                                 | Gear efficiency                         |
| $r_w$           | m                                                 | Dynamic wheel radius                    |
| $k_c, k_i, k_w$ | –                                                 | Copper, iron, windage loss coefficients |
| $P_0$           | W                                                 | Constant electronics loss               |
| $V_{oc}$        | V                                                 | Open-circuit voltage                    |
| $R_i$           | $\Omega$                                          | Pack internal resistance                |
| $I$             | A                                                 | Pack current                            |
| $z$             | –                                                 | State of charge                         |
| $q$             | A h                                               | Pack charge                             |

## 2 Reference Vehicle and Parameter Set

### 2.1 Vehicle Description

The Model 3 RWD is a rear-wheel-drive saloon with a single interior permanent-magnet synchronous machine driving the rear axle through a single-speed reduction gear. The 2024 Highland revision carries a lithium-iron-phosphate (LFP) battery, a chemistry whose open-circuit voltage curve is characteristically flat across the middle of its state-of-charge range. Table 2 lists the parameters the model requires, grouped by subsystem, with every estimated value shown in red.

### 2.2 Parameter Provenance

Values are drawn from the manufacturer’s own documentation where it exists [3], from the procedures that type approval prescribes [4, 5] where a measured value is needed but not published, and from the standard automotive literature [6, 7] otherwise. Where a quantity could not be traced to any of those, it is an estimate and is marked as one. The distinction matters and is recorded for every entry in

the machine-readable parameter file that accompanies the software. Figure 1 summarises the split.

The published performance figures listed at the foot of Table 2 are used *only* for validation. They are never model inputs, so the comparison in Section 8 remains honest.

Table 2: Parameter set. Values shown in red are engineering estimates rather than published data, as the project brief requires.

| Parameter                                          | Unit                   | Value   |
|----------------------------------------------------|------------------------|---------|
| <i>Mass and geometry</i>                           |                        |         |
| Kerb mass                                          | kg                     | 1765    |
| Test payload                                       | kg                     | 100     |
| Rotational mass factor $\lambda$                   | –                      | 1.04    |
| Centre-of-gravity height $h$                       | m                      | 0.46    |
| Wheelbase $L$                                      | m                      | 2.875   |
| Rear static load fraction $\chi$                   | –                      | 0.52    |
| <i>Aerodynamics and tyres</i>                      |                        |         |
| Drag coefficient $C_d$                             | –                      | 0.219   |
| Frontal area $A$                                   | m <sup>2</sup>         | 2.22    |
| Air density $\rho$                                 | kg/m <sup>3</sup>      | 1.204   |
| Dynamic wheel radius $r_w$                         | m                      | 0.3323  |
| Rolling resistance $f_{r0}$                        | –                      | 0.0080  |
| Peak friction $\mu$                                | –                      | 0.85    |
| <i>Road load (coastdown)</i>                       |                        |         |
| $F_0$                                              | N                      | 115.65  |
| $F_1$                                              | N/(m/s)                | 1.8906  |
| $F_2$                                              | N/(m/s) <sup>2</sup>   | 0.37836 |
| Coastdown test mass                                | kg                     | 1900    |
| <i>Motor and transmission</i>                      |                        |         |
| Peak power $P_{\max}$                              | kW                     | 208     |
| Peak torque $T_{\max}$                             | N m                    | 420     |
| Maximum motor speed                                | rpm                    | 17 900  |
| Gear ratio $i_g$                                   | –                      | 9.0     |
| Gear efficiency $\eta_g$                           | –                      | 0.98    |
| Copper loss $k_c$                                  | W/(N m) <sup>2</sup>   | 0.0904  |
| Iron loss $k_i$                                    | W/(rad/s)              | 0.669   |
| Windage loss $k_w$                                 | W/(rad/s) <sup>3</sup> | 1.79e-6 |
| Constant loss $P_0$                                | W                      | 185.9   |
| <i>Battery</i>                                     |                        |         |
| Gross capacity                                     | kWh                    | 62.5    |
| Usable fraction                                    | –                      | 0.96    |
| Nominal voltage                                    | V                      | 345     |
| Internal resistance $R_i$                          | $\Omega$               | 0.075   |
| Max. discharge power                               | kW                     | 220     |
| Max. charge power                                  | kW                     | 70      |
| <i>Auxiliaries</i>                                 |                        |         |
| Base auxiliary load                                | W                      | 300     |
| HVAC load (certification)                          | W                      | 0       |
| <i>Published figures, used for validation only</i> |                        |         |
| 0–100 km/h                                         | s                      | 6.1     |
| Top speed                                          | km/h                   | 201     |
| WLTP range                                         | km                     | 513     |

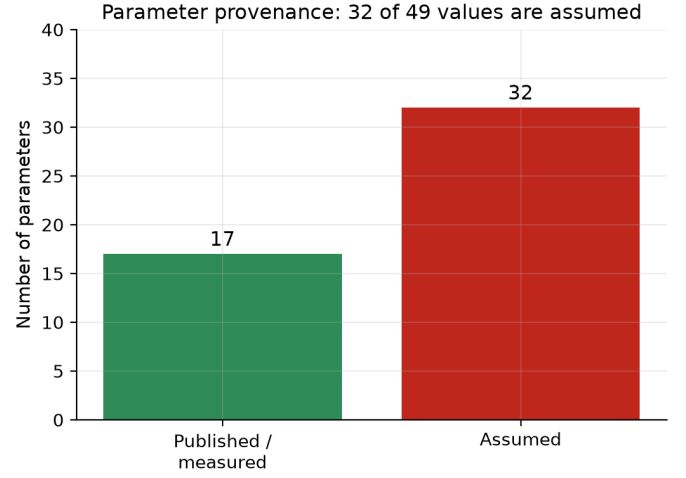


Figure 1: Provenance of the parameter set. Thirty-two of forty-nine values are engineering estimates, because the manufacturer publishes no motor map, gear ratio or usable battery capacity.

## 3 Longitudinal Vehicle Model

### 3.1 Force Balance

The force the vehicle must deliver at the wheels to follow a demanded speed  $v$  and acceleration  $a$  is

$$F_{\text{trac}} = F_{\text{inertia}} + F_{\text{run}}(v) + F_{\text{grade}}, \quad (1)$$

where the inertia term accounts for the rotating masses of wheels, shafts and rotor through a factor  $\lambda$ ,

$$F_{\text{inertia}} = \lambda m a, \quad F_{\text{grade}} = m g \sin \alpha. \quad (2)$$

### 3.2 Running Resistance: Two Formulations

Two formulations of the running resistance  $F_{\text{run}}$  are implemented, and the difference between them is a result in its own right.

#### 3.2.1 Textbook Decomposition

The decomposition taught in the standard texts [6, 7] separates the two physical mechanisms,

$$F_{\text{run}}^{\text{phys}}(v) = \underbrace{f_r(v) m g \cos \alpha}_{\text{rolling}} + \underbrace{\frac{1}{2} \rho C_d A v^2}_{\text{aerodynamic}}, \quad (3)$$

with a speed-dependent rolling coefficient  $f_r(v) = f_{r0} + kv^2$ . This form is what the report derives, and it is what the traction diagram of Section 4 plots, because it shows which mechanism dominates where.

### 3.2.2 Regulatory Coastdown Form

It is not, however, what the model uses to predict energy. Type approval does not measure rolling resistance and aerodynamic drag separately. It rolls the vehicle down from speed on a level test track and fits a quadratic [4, 5],

$$F_{\text{run}}^{\text{cd}}(v) = F_0 \frac{m}{m_{\text{cd}}} + F_1 v + F_2 v^2. \quad (4)$$

This coastdown form is the default in the model because it is physically the more complete description, for two reasons.

First, the linear term  $F_1$  captures bearing, seal and driveline drag, for which Equation 3 has no counterpart at all. An electric drive unit cannot be disconnected during a coastdown, so its spin losses are necessarily inside the measured coefficients.

Second, the quadratic term captures the aerodynamic drag the vehicle actually experiences on the road, including wheel rotation and cooling airflow, both of which a wind-tunnel drag coefficient excludes by construction.

The constant term  $F_0$  is tyre rolling resistance, which is proportional to normal load, so it is scaled by the ratio of the actual mass  $m$  to the mass  $m_{\text{cd}}$  at which the coastdown was performed. Without that correction the model reports payload as very nearly free, because the kinetic energy that extra mass adds is largely returned by regenerative braking.

### 3.2.3 Comparison

Table 3 compares the two. The published drag coefficient of  $C_d = 0.219$  implies a drag area  $C_d A = 0.486 \text{ m}^2$ , whereas the coastdown quadratic implies  $0.629 \text{ m}^2$ , higher by 29%. Both numbers are correct; they measure different things. Using Equation 3 alone under-predicts the running resistance by 12 to 18% above 80 km/h, and the energy demand by a comparable margin.

Table 3: Running resistance from the two formulations. The textbook decomposition has no linear term and uses a wind-tunnel drag coefficient, so it falls progressively short as speed rises.

| Speed<br>km/h | Coastdown<br>N | Textbook<br>N | Difference<br>% |
|---------------|----------------|---------------|-----------------|
| 30            | 155.5          | 168.2         | +8.1            |
| 50            | 212.8          | 207.0         | -2.7            |
| 80            | 342.4          | 301.7         | -11.9           |
| 100           | 458.0          | 389.1         | -15.0           |
| 130           | 675.2          | 556.6         | -17.6           |
| 150           | 849.2          | 692.5         | -18.4           |

## 3.3 Powertrain

### 3.3.1 Torque and Power Envelope

A traction machine is operated in two regions. Below the corner point the current limit binds and the torque is constant; above it the voltage limit binds and the machine enters field weakening at constant power [2],

$$T_{\text{max}}(\omega) = \begin{cases} T_{\text{pk}}, & \omega \leq \omega_b, \\ \frac{P_{\text{pk}}}{\omega}, & \omega_b < \omega \leq \omega_{\text{max}}, \end{cases} \quad (5)$$

with the corner point at  $\omega_b = P_{\text{pk}}/T_{\text{pk}}$ . For this vehicle  $\omega_b$  corresponds to 4729 rpm, which is 65.8 km/h at the wheels. The tractive force available at the road is

$$F_{\text{mot}}(v) = \frac{T_{\text{max}}(\omega) i_g \eta_g}{r_w}, \quad \omega = \frac{v i_g}{r_w}. \quad (6)$$

### 3.3.2 Loss Model

A single fixed efficiency is not adequate here. A legislative cycle spends most of its time at light load, where a real drive unit is markedly less efficient than at its rated point, so a constant value would misstate the cycle energy in whichever direction it was calibrated. The combined motor-and-inverter loss is therefore represented by the four-term model of Larminie and Lowry [8],

$$P_{\text{loss}} = k_c T^2 + k_i \omega + k_w \omega^3 + P_0, \quad (7)$$

whose terms are respectively copper (current-dependent), iron (flux-dependent), windage, and constant electronics loss.

The four coefficients were fitted by least squares to seven operating points typical of a liquid-cooled automotive permanent-magnet machine: approximately 97% peak efficiency, 92.5% at peak power, 88% at light load and high speed corresponding to motorway cruise, and 80% at light load and low speed. The resulting map is shown in Figure 5. All four coefficients are **assumed**.

## 3.4 Battery

The pack is modelled as a state-of-charge dependent open-circuit voltage in series with a constant internal resistance, the *Rint* equivalent circuit [9, 10],

$$P_{\text{term}} = V_{\text{oc}}(z) I - I^2 R_i, \quad (8)$$

which is solved for the current drawn at a demanded terminal power,

$$I = \frac{V_{\text{oc}} - \sqrt{V_{\text{oc}}^2 - 4R_i P_{\text{term}}}}{2R_i}. \quad (9)$$

The radicand turns negative beyond the resistance-limited power ceiling  $V_{oc}^2/4R_i$ , which the implementation reports as a limit rather than silently returning a complex value.

State of charge is tracked in ampere-hours rather than in energy. This matters: it charges the ohmic loss against the pack instead of ignoring it, and counts it correctly twice on a cycle with heavy regeneration, once on the way out and once on the way back in. The pack capacity is sized so that  $\int V_{oc} dq$  across the usable window equals the nameplate usable energy, giving 175.3 Ah at a window-averaged open-circuit voltage of 342.3 V. A slow discharge then returns 59.96 kWh at the terminals against a 60.0 kWh nameplate, the difference being the ohmic loss.

The LFP open-circuit voltage curve is **assumed**, scaled from published cell data. It rises only about 6 % between 25 % and 85 % state of charge, which is the characteristic flat plateau of the chemistry.

### 3.5 Simulation Method

The simulation is quasi-static and backward-facing, the standard approach for energy and range work [1]: the speed trace is the input, and the model asks what the powertrain must do to follow it. No driver model is required. Within each one-second step the forces are evaluated at the mid-point speed

$$\bar{v} = \frac{1}{2} (v_k + v_{k+1}), \quad a_k = \frac{v_{k+1} - v_k}{\Delta t}, \quad (10)$$

which makes the integration second-order rather than first-order accurate.

Every limit, mechanical and electrical, is resolved *before* the step is committed. If the motor envelope, the tyre adhesion limit, the battery power rating or the remaining charge cannot deliver what the trace demands, the model substitutes the acceleration that is actually achievable and flags the step. It never reports a vehicle following a trace on energy the battery did not supply. Over the WLTC the vehicle meets the demand at every one of the 1800 steps, with a peak tracking error of 0 km/h.

## 4 Speed and Acceleration Characteristics

### 4.1 Tyre Adhesion Limit

The force available at the road is the lesser of what the machine can produce and what the driven tyres can transmit. For a rear-wheel-drive vehicle the second of these binds first, and the problem is implicit: accelerating transfers load onto the driven axle, which raises the adhesion limit, which permits more acceleration. The normal load on the rear axle

is

$$N_r = \chi m g \cos \alpha + \frac{m a h}{L}, \quad (11)$$

the adhesion limit is  $F_{adh} = \mu N_r$ , and the acceleration is  $a = (F - F_{res})/(\lambda m)$ . Eliminating  $a$  gives a closed form, which the implementation solves directly rather than iterating:

$$F_{adh} = \frac{\mu \chi m g \cos \alpha - \frac{\mu h F_{res}}{L \lambda}}{1 - \frac{\mu h}{L \lambda}}. \quad (12)$$

This is not a refinement of secondary importance. At rest the tyres can transmit 9.30 kN while the motor could deliver 11.15 kN, so the vehicle is traction-limited up to 79 km/h, which is 77 % of the time it takes to reach 100 km/h.

Both ways of simplifying the tyre limit fail, and they fail in opposite directions. Ignoring it altogether leaves only the torque limit and gives 5.41 s, 12 % optimistic. Using the static axle load without the transfer term gives 6.90 s, 12 % pessimistic. Only the implicit solution of Equation 12, which lets the load transfer raise the limit as the vehicle accelerates, lands on the published figure.

### 4.2 Traction Diagram

Figure 2 plots the available force against the running resistance. The motor envelope is flat to the corner point at 65.8 km/h and falls as  $1/v$  thereafter. The adhesion limit, drawn in the assumption colour because it depends on an estimated friction coefficient, cuts below the envelope up to 79 km/h. The dashed curve is the textbook decomposition of Equation 3, shown for comparison; the gap between it and the solid total resistance is the shortfall quantified in Table 3.

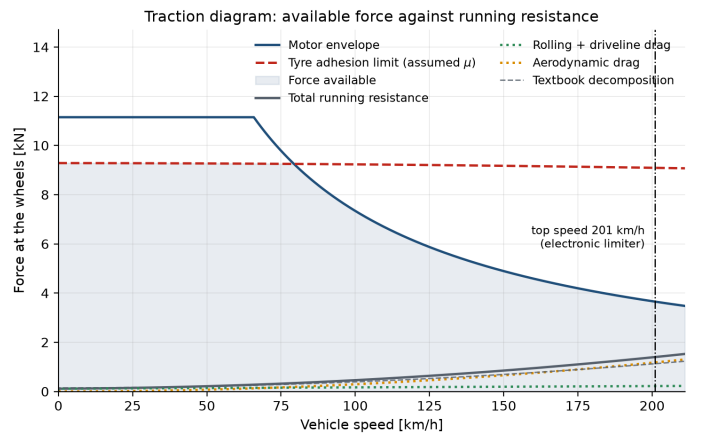


Figure 2: Traction diagram. Available force against running resistance, with the tyre adhesion limit and the textbook road-load decomposition shown for comparison.

### 4.3 Acceleration Characteristic

The maximum acceleration at each speed follows directly,

$$a_{\max}(v) = \frac{\min[F_{\text{mot}}(v), F_{\text{adh}}(v)] - F_{\text{run}}(v)}{\lambda m}. \quad (13)$$

The full-throttle launch is obtained by integrating

$$t = \int_0^v \frac{du}{a_{\max}(u)}, \quad (14)$$

on a fine speed grid, which is numerically far better behaved near standstill than stepping in time. Figure 3 shows both, with the tyre-limited region shaded on the left panel and the benchmark times marked on the right. Table 4 collects the results.

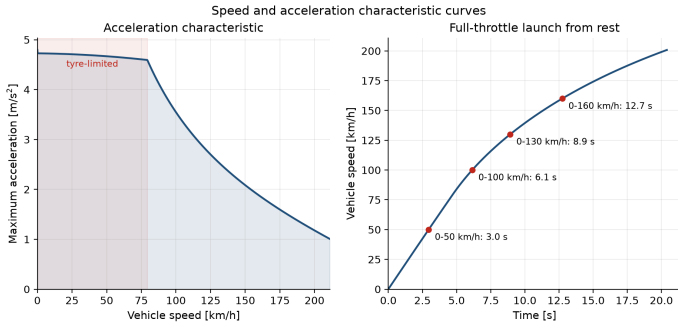


Figure 3: Speed and acceleration characteristic curves. Left: maximum acceleration against speed, with the tyre-limited region shaded. Right: the full-throttle launch from rest.

Table 4: Predicted performance characteristics.

| Quantity                       | Value | Unit             |
|--------------------------------|-------|------------------|
| Maximum acceleration from rest | 4.79  | m/s <sup>2</sup> |
| Maximum tractive force         | 9.30  | kN               |
| Maximum wheel power            | 203.8 | kW               |
| 0–50 km/h                      | 2.95  | s                |
| 0–100 km/h                     | 6.14  | s                |
| 0–130 km/h                     | 8.93  | s                |
| 0–160 km/h                     | 12.74 | s                |
| Top speed                      | 201   | km/h             |

### 4.4 Top Speed and Gradeability

Three distinct constraints can cap the top speed of a battery-electric vehicle, and which one binds is a design statement worth reporting rather than a number to be quoted in isolation. They are the tractive force balance, the maximum mechanical speed of the machine, and the electronic limiter.

For this vehicle the limiter binds, at 201 km/h. Removing it, the machine reaches its **17 900 rpm** ceiling at 249 km/h with tractive force still in reserve, so the force balance never binds at all. The implementation reports which of the three is active.

Gradeability follows from setting the acceleration to zero and solving the force balance for the slope. Figure 4 shows the power envelope and the maximum climbable gradient against speed. The vehicle will pull away on a gradient well beyond anything found on a public road, which is characteristic of electric traction and its full torque from rest.

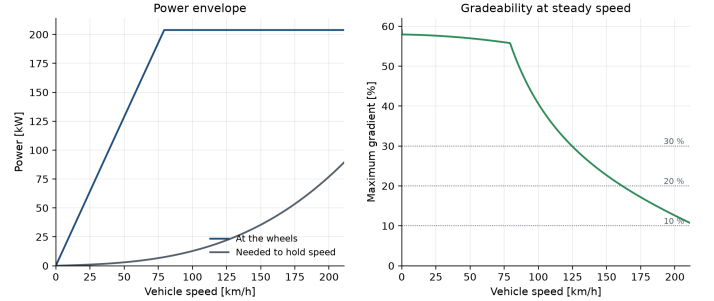


Figure 4: Power envelope at the wheels and maximum climbable gradient at steady speed.

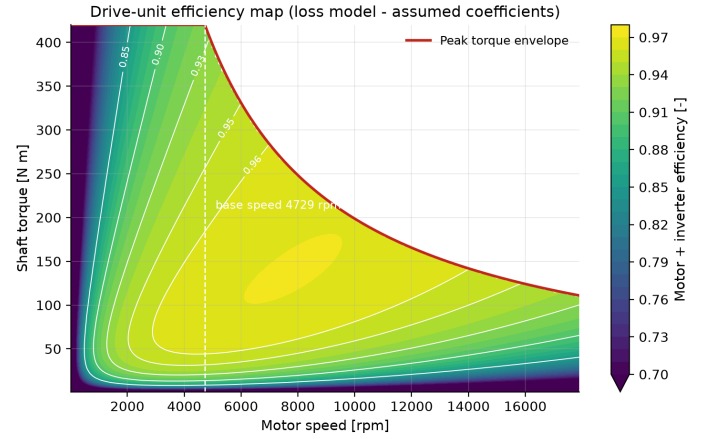


Figure 5: Combined motor and inverter efficiency map from Equation 7, with the peak torque envelope overlaid. The coefficients are **assumed**, fitted to seven representative operating points.

## 5 Driving Profile

### 5.1 Choice of Cycle

The WLTC Class 3b cycle was selected because the vehicle's own published range is certified on it [4], so the model and its reference describe the same drive. Comparing a prediction made on one profile against a figure certified on another would confound the model error with the difference between the cycles.

The cycle runs for 1800 s over 23.27 km in four phases of rising speed, labelled Low, Medium, High and Extra-high, reaching a maximum of 131.3 km/h. Figure 6 shows the speed and acceleration traces with the phases marked.

## 5.2 Profile Synthesis

The official second-by-second table of the WLTC is not reproduced here. Instead the profile is **synthesised** from the published phase statistics. For each phase the construction specifies a sequence of speed levels, ramps linearly between them at realistic rates, and solves for the hold durations. Two linear constraints on the  $n$  non-zero hold durations  $t$  fix the phase duration and its distance,

$$\sum_i t_i = T_{\text{drive}} - T_{\text{ramp}}, \quad \sum_i v_i t_i = D - D_{\text{ramp}}, \quad (15)$$

which for  $n > 2$  is underdetermined. The solution closest to a nominal distribution is taken by solving the equality-constrained least-squares problem  $\min \|t - t_0\|^2$  subject to  $At = b$  through its Lagrangian,  $t = t_0 + A^T(AA^T)^{-1}(b - At_0)$ .

The resulting trace is not the official one second by second, but every published statistic is reproduced to within half a per cent, as Table 5 shows. More importantly, the energy result does not depend on the shape of the synthetic trace: sweeping the acceleration rates across their whole plausible span moves the predicted consumption by only 1.9%. Supplying the official table overrides the synthesis automatically.

Table 5: Synthesised WLTC Class 3b against the published statistics.

| Statistic             | Generated | Published | Dev.    |
|-----------------------|-----------|-----------|---------|
| Duration (s)          | 1800.0    | 1800.0    | +0.00 % |
| Distance (km)         | 23.268    | 23.266    | +0.01 % |
| Maximum speed (km/h)  | 131.30    | 131.30    | +0.00 % |
| Mean speed (km/h)     | 46.54     | 46.50     | +0.08 % |
| Stop time (s)         | 243.0     | 242.0     | +0.41 % |
| Low phase (km)        | 3.096     | 3.095     | +0.02 % |
| Medium phase (km)     | 4.756     | 4.756     | +0.00 % |
| High phase (km)       | 7.162     | 7.162     | −0.00 % |
| Extra-high phase (km) | 8.254     | 8.254     | −0.00 % |

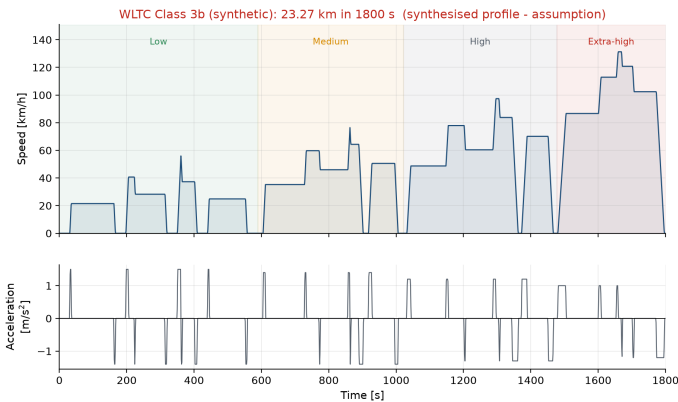


Figure 6: WLTC Class 3b speed and acceleration traces, with the four phases shaded.

## 5.3 Comparison Cycle

The New European Driving Cycle (NEDC) is also implemented, reconstructed directly from its regulatory segment table [11]. Because that table specifies the cycle as idle, constant-acceleration and constant-speed segments, the reconstruction is deterministic. It yields 1180 s and 10.975 km against the official 11.007 km, low by 0.29%. The extra-urban half reproduces its official 6.955 km exactly, so the whole shortfall sits in the urban half. In the segment list used here those durations sum to 194 s rather than the specified 195 s, and the missing second is carried in the final idle; whether that reflects a transcription error or an ambiguity in the source table was not established.

The NEDC serves to show how much the choice of profile alone changes the answer, and is discussed in Section 8. Its much gentler speed and acceleration content, visible in Figure 7, is the reason it was replaced by the WLTP [12, 13]. Where the WLTC is built from irregular micro-trips with continuously varying speed, the NEDC consists of repeated trapezoidal segments at constant acceleration, a profile no driver produces and one that understates both the transient losses and the high-speed drag a vehicle actually meets.

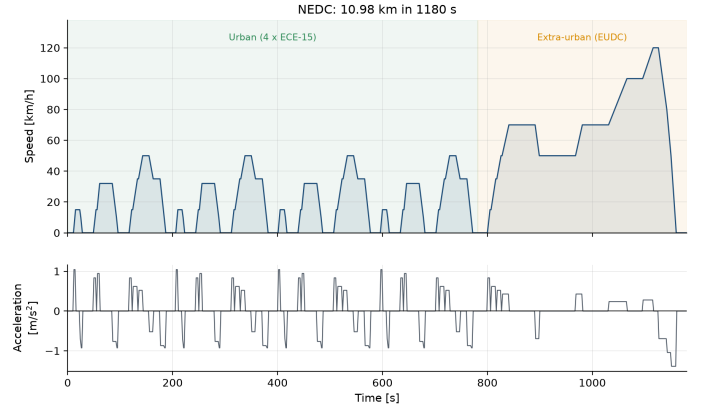


Figure 7: NEDC speed and acceleration traces, reconstructed from the regulatory segment table. The four repeated urban segments are followed by a single extra-urban phase.

## 6 Cycle Simulation and Energy Balance

### 6.1 Results Over One Cycle

Over one pass of the WLTC the vehicle covers 23.27 km and draws 2.665 kWh net from the battery, giving a consumption of 11.45 kWh/100 km at the terminals. Regenerative braking returns 24.6% of the traction energy. Figure 8 shows the speed, battery power and state of charge over the cycle. The demanded trace is entirely hidden behind the achieved

one, which is the point: the powertrain follows the profile at every step.

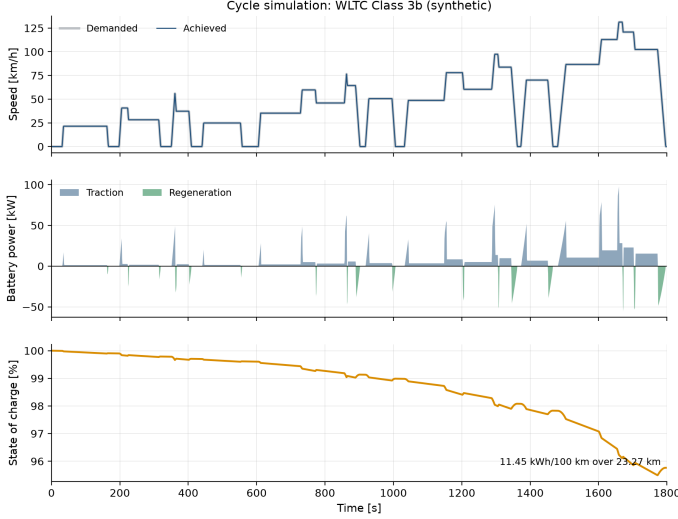


Figure 8: Cycle simulation. Demanded and achieved speed, battery power split into traction and regeneration, and state of charge.

## 6.2 Energy Demand by Phase

Because the four WLTC phases differ so widely in speed, the cycle can be read as four separate drives. Table 6 reports the consumption of each.

The specific consumption rises from 81.5 Wh/km in the Low phase to 156.0 Wh/km in the Extra-high phase, very nearly a doubling. Two mechanisms drive this. Aerodynamic drag grows with the square of speed and dissipates energy that cannot be recovered, and the share of traction energy returned by regeneration falls from 27.7 % to 17.0 % because a high-speed phase spends proportionally less of its time braking.

The consequence for a driver is direct: the Extra-high phase contributes 35 % of the cycle distance but 48 % of its energy. A journey conducted entirely at motorway speed will fall well short of the certified range even before any auxiliary load is considered, and Figure 11 quantifies that case.

Table 6: Energy demand by WLTC phase. Specific consumption nearly doubles between the Low and Extra-high phases.

| Phase        | Dist.<br>km   | $v_{avg}$<br>km/h | Energy<br>kWh | Spec.<br>Wh/km | Regen<br>%  |
|--------------|---------------|-------------------|---------------|----------------|-------------|
| Low          | 3.096         | 18.9              | 0.2524        | 81.5           | 27.7        |
| Medium       | 4.756         | 39.5              | 0.3959        | 83.2           | 30.0        |
| High         | 7.162         | 56.7              | 0.7286        | 101.7          | 27.9        |
| Extra-high   | 8.254         | 92.0              | 1.2878        | 156.0          | 17.0        |
| <b>Total</b> | <b>23.268</b> | <b>46.5</b>       | <b>2.6646</b> | <b>114.5</b>   | <b>24.6</b> |

## 6.3 Energy Balance

Table 7 decomposes where the energy goes. The four terms close against the net battery energy to  $1.3 \times 10^{-7}$  J, which is a check on the book-keeping rather than a coincidence: the drive-unit loss is measured directly as the difference between electrical and wheel power in each step, not inferred by subtracting the other terms.

The road load dominates at 77 % of the total. Drive-unit losses account for 17 %, the auxiliary load for 5.6 %, and friction braking for only 0.3 %, the last because the cycle’s decelerations are gentle enough that the motor absorbs nearly all of them within its regeneration limit. Figure 9 shows the same decomposition alongside the flow of energy into and out of the battery.

Table 7: Energy balance over one WLTC cycle. The four expenditure terms close exactly against the net battery energy.

| Term                                 | kWh           | Share        |
|--------------------------------------|---------------|--------------|
| Road load (rolling, driveline, drag) | 2.0608        | 77.3 %       |
| Drive-unit losses                    | 0.4450        | 16.7 %       |
| Auxiliaries                          | 0.1500        | 5.6 %        |
| Friction brakes                      | 0.0088        | 0.3 %        |
| <b>Net from the battery</b>          | <b>2.6646</b> | <b>100 %</b> |

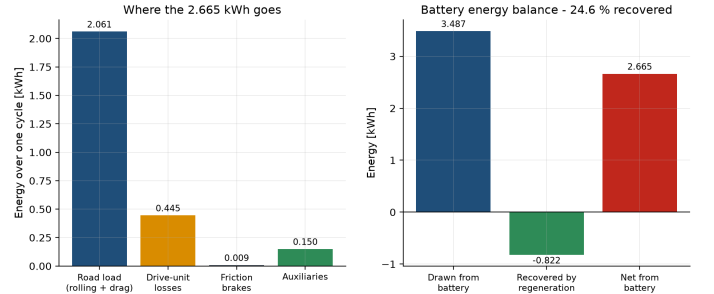


Figure 9: Energy expenditure over one cycle and the battery energy balance, showing the share recovered by regeneration.

## 7 Range on a Full Battery

### 7.1 Two Independent Methods

Range is computed two ways, which serves as a cross-check on the energy accounting.

The *analytic* method divides the usable battery energy by the consumption per kilometre. The *integrated* method repeats the cycle, carrying the state of charge forward, until the usable window is exhausted, so that it sees the ohmic loss grow as the open-circuit voltage falls towards the end of the discharge.

Both divisions must be performed at the open-circuit level, because that is where the usable energy is defined: coulomb counting makes the state-of-charge window hold  $\int V_{oc} dq$ , not  $\int P_{term} dt$ . Dividing the usable energy by the terminal consumption instead would overstate the range by roughly the ohmic loss, and that unit error would then masquerade as a disagreement between the two methods.

With both on the same footing they agree to 0.64 %, the integrated method giving the shorter range, which is the direction the physics demands. Table 8 gives the figures and Figure 10 the discharge trace.

Table 8: Range on a full battery over the WLTC Class 3b cycle.

| Quantity                     | Value        | Unit      |
|------------------------------|--------------|-----------|
| Range, integrated method     | 508.3        | km        |
| Range, analytic method       | 511.6        | km        |
| Disagreement between methods | −0.64        | %         |
| Consumption at the terminals | 114.5        | Wh/km     |
| Consumption at open circuit  | 117.3        | Wh/km     |
| Usable battery energy        | 60.0         | kWh       |
| Cycle repetitions to empty   | 21.8         | –         |
| <b>Published WLTP range</b>  | <b>513.0</b> | <b>km</b> |
| <b>Deviation</b>             | <b>−0.92</b> | <b>%</b>  |

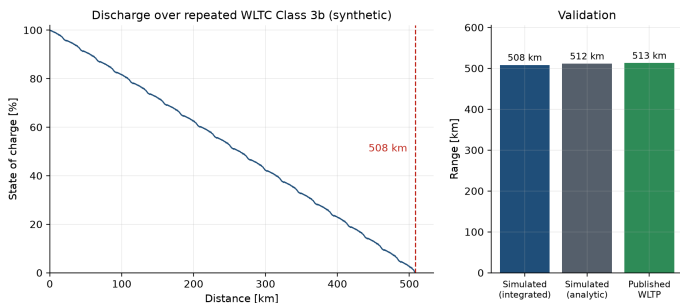


Figure 10: State of charge against distance over repeated cycles to an empty battery, and comparison of both methods against the published figure.

## 7.2 Range at Constant Speed

Figure 11 shows range and consumption at steady cruising speed, a classic electric-vehicle characteristic. The curve has an interior maximum because the constant auxiliary load dominates at low speed, where little distance is covered per unit time, while aerodynamic drag dominates at high speed. The predicted optimum lies near 28 km/h.

This curve is also an independent validation, discussed in the next section, because none of its points was used in any calibration.

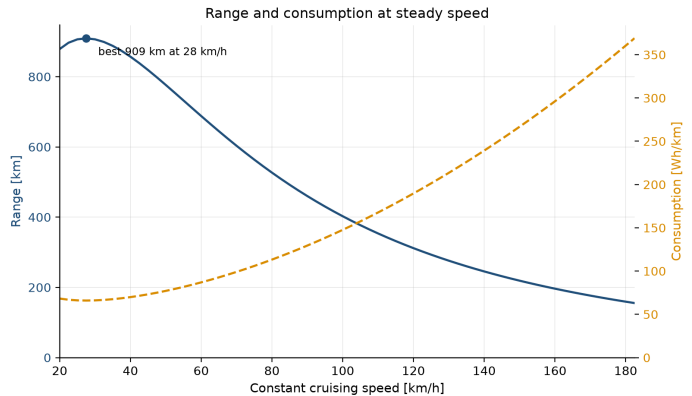


Figure 11: Range and consumption at steady cruising speed. The interior maximum arises from the competition between the constant auxiliary load and aerodynamic drag.

## 8 Validation

### 8.1 Against Published Figures

Table 9 compares the model against the three figures the manufacturer publishes. All three are reproduced from a single parameter set, and none of them was used as a model input.

That claim deserves to be stated precisely, because the distinction between checking a model and fitting one is easy to blur. Every parameter was set from an independent source before the comparison was made: published data where it exists, certification coastdown targets for the road load, and efficiency targets drawn from the machines literature for the loss model. The published acceleration, top speed and range were then used to test the result. Where a deviation appeared during development it was resolved by correcting the physics rather than by adjusting a parameter until the figure matched. The two corrections that mattered are described in Section 3: the drive-unit loss model originally made windage dominant at cruise, and the textbook road-load decomposition omitted the driveline term entirely. Both were identified from their own internal implausibility, not from the size of the range error.

Table 9: Validation against published manufacturer data. No published figure was used as a model input.

| Quantity                 | Sim.  | Pub.  | Dev.    | Tol.  |
|--------------------------|-------|-------|---------|-------|
| 0–100 km/h (s)           | 6.14  | 6.10  | +0.68 % | ±10 % |
| Top speed (km/h)         | 201.0 | 201.0 | +0.00 % | ±5 %  |
| WLTP range (km)          | 508.3 | 513.0 | −0.92 % | ±10 % |
| Consumption (kWh/100 km) | 11.45 | 11.70 | −2.08 % | ±10 % |

### 8.2 Independent Checks

Three further checks were applied that use no manufacturer data at all.

*Steady-cruise consumption.* The model predicts 148 Wh/km at 100 km/h and 213 Wh/km at 130 km/h. Neither point entered any calibration: both follow from the same road-load and drivetrain parameters that were fixed before the cycle was run. Their magnitude is consistent with the specific energy consumption reported for battery-electric vehicles in real-world driving, which for a compact vehicle monitored over four years varied widely with ambient temperature, traffic and road grade [14]. No vehicle-specific measured figure for the Model 3 at these speeds was available to the authors, so this is offered as a plausibility check rather than as a validation against a certified value.

*Energy conservation at the wheel.* Integrating Newton’s second law against speed over a cycle that starts and ends at rest requires the net wheel work to equal the road-load dissipation plus the friction-brake dissipation. The residual is  $9 \times 10^{-10}$  J.

*Closure of the battery energy balance.* The four expenditure terms of Table 7 close against the net battery energy to  $1 \times 10^{-7}$  J, and the loop accumulators agree with the integral of the battery power trace to the same tolerance under every limiting condition tested.

### 8.3 Effect of the Driving Profile

Run over the NEDC instead, the same vehicle returns 10.90 kWh/100 km and 541 km, which is 6.4 % further than on the WLTC. The older cycle is gentler in both speed and acceleration content and therefore flatters the vehicle, which is precisely the reason it was replaced.

## 9 Assumptions and Sensitivity

### 9.1 Rationale

Thirty-two of the forty-nine parameters are estimates. Listing them, as Table 2 does, is necessary but not sufficient. What matters for the credibility of the result is how much each one moves the answer.

Table 10 varies each parameter across the interval a reasonable engineer would call plausible and reports the resulting span in predicted range. Figure 12 shows the same information as a tornado chart.

An important subtlety is that the set of parameters which influence the result depends on which road-load formulation is active. Under the coastdown form of Equation 4 the wind-tunnel drag coefficient, the frontal area and the tyre rolling-resistance coefficient influence nothing at all. A sensitivity study that varied them regardless would place silent zeros on the chart, which would look convincing and mean nothing. The case list is therefore built from the active model.

Table 10: Sensitivity of the predicted range to each assumed parameter, ranked by influence.

| Parameter           | Interval    | Span<br>km | Of base<br>% |
|---------------------|-------------|------------|--------------|
| HVAC load           | 0–3000 W    | 181        | 35.6         |
| Regen power limit   | 0–90 kW     | 115        | 22.6         |
| Battery capacity    | 57.5–66 kWh | 70         | 13.7         |
| Coastdown $F_2$     | 0.340–0.440 | 52         | 10.1         |
| Coastdown $F_0$     | 95–140 N    | 52         | 10.1         |
| Payload             | 0–400 kg    | 40         | 7.9          |
| Coastdown $F_1$     | 1.30–2.45   | 28         | 5.6          |
| Gear efficiency     | 0.960–0.992 | 25         | 4.8          |
| Motor windage $k_w$ | 1.0–3.0 e–6 | 22         | 4.3          |

### 9.2 Basis of the Dominant Assumptions

Because the ranking in Table 10 identifies which estimates carry the result, the basis of the five most influential is stated explicitly in Table 11. The remaining assumptions are documented in the same form in the machine-readable parameter file.

Table 11: Basis for the five assumptions that most influence the predicted range. All values are **estimates**.

| Assumption                                         | Basis                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HVAC load<br><b>0–3000 W</b>                       | Certification is run with auxiliaries off. The upper bound corresponds to cabin and battery heating at 0 °C, consistent with published cold-weather consumption studies [15].                                                                                                                                                  |
| Regen power limit<br><b>70 kW</b>                  | Observed regeneration ceiling on single-motor variants. The lower bound of the interval represents friction braking only.                                                                                                                                                                                                      |
| Gross capacity<br><b>62.5 kWh</b>                  | Not published. Taken from independent pack teardowns and battery-management logs for the LFP pack; the interval spans the range those sources report.                                                                                                                                                                          |
| Coastdown $F_0$ ,<br>$F_1$ , $F_2$                 | Coefficients of the form certification prescribes [5], converted to SI. The values are representative of this vehicle class rather than read from a certification record for this exact variant, which is why all three are estimates; the intervals reflect the variation with wheel size, tyre specification and model year. |
| Loss coefficients<br>$k_c$ , $k_i$ , $k_w$ , $P_0$ | Fitted by least squares to seven operating points representative of a liquid-cooled automotive permanent-magnet drive unit [2, 8].                                                                                                                                                                                             |

### 9.3 Interpretation

The ranking is itself a result. The auxiliary load dominates, worth 35.6 % of the baseline range on its own, and it is the one quantity in the list that the driver actually controls.

The regeneration power limit follows, at 22.6 %.

The drivetrain assumptions, which were the hardest to pin down and which required the loss-model fit described in Section 3, matter least: gear efficiency and motor windage together account for under 10 %. This is the reassuring outcome. The answer does not rest on the values that were estimated with the least confidence.

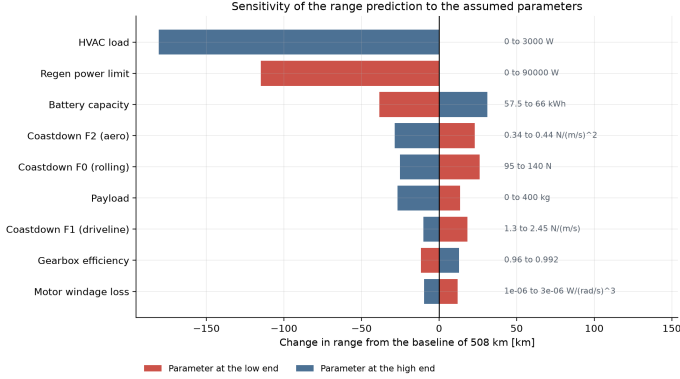


Figure 12: Sensitivity of the predicted range to each assumed parameter across its plausible interval.

## 10 Real-World Conditions

The certification figure is measured with the auxiliaries switched off, a defined test mass and standard air. Real driving is none of those things [15]. Table 12 and Figure 13 give the predicted range under five named conditions, all evaluated on the same cycle and differing only in the parameters a driver or the weather changes.

The spread between the certification case and a winter morning with cabin and battery heating is 215 km, or 42 %. A driver told to expect 508 km and finding 293 km has not been misled by a faulty vehicle; they have encountered the difference between a standardised test and the world.

Table 12: Predicted range under real-world conditions, all on the WLTC cycle.

| Scenario           | Conditions               | Range km | $\Delta$ % |
|--------------------|--------------------------|----------|------------|
| WLTP certification | Aux. off, driver only    | 508      | —          |
| Mild weather       | Ventilation, 2 occupants | 455      | -10        |
| Loaded, roof box   | 5 occupants plus luggage | 394      | -22        |
| Summer, A/C        | 30 °C ambient            | 378      | -26        |
| Winter, heating    | 0 °C, cold tyres         | 293      | -42        |

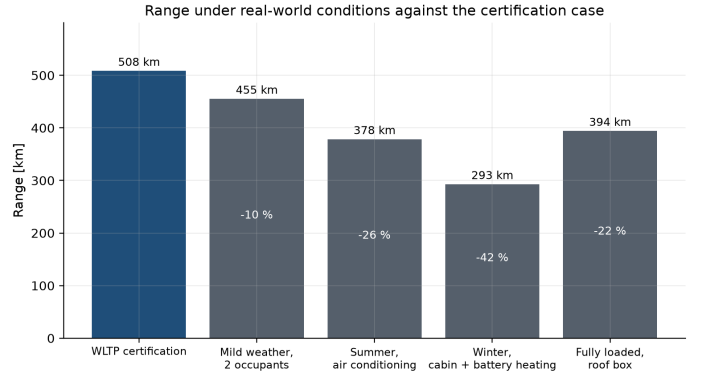


Figure 13: Predicted range under real-world conditions against the certification case.

## 11 Software Implementation

### 11.1 Structure

The model is implemented as an installable Python package of fifteen modules, built on NumPy [16], SciPy [17] and Matplotlib [18]. Responsibilities are separated by physical subsystem: parameter loading and the assumption register, road load, powertrain, battery, performance characteristics, driving cycles, the cycle simulator, range calculation, the sensitivity study, plotting, and report generation.

All internal computation is in SI units, with conversion confined to the input and output boundaries. Every parameter is loaded from a single versioned YAML file in which each entry carries its numeric value, its unit, a provenance string, and a boolean flag recording whether it is an assumption. The red marking in this report is generated from that flag rather than applied by hand, so it cannot fall out of step with the model.

### 11.2 Reproducibility and Verification

A single command regenerates every figure, table and document in this report from the parameter file alone, taking approximately 80 s. The numbers quoted in the text are produced by the same run that produces the figures, so they cannot diverge.

The package is accompanied by 125 automated tests, which execute in about 76 s. They check each road-load term against an independent hand calculation rather than against the code’s own output; verify that the motor envelope corner point, the constant-power region and the peak efficiency fall in physically plausible bands; confirm that a slow full discharge returns the nameplate energy and that the current solution satisfies Equation 9; compare both driving cycles against their published statistics phase by phase; and assert the acceptance criteria stated in Section 8.

Two tests deserve specific mention because they guard

against failures that would otherwise be invisible. One drives an artificially underpowered vehicle over the cycle and asserts that the simulator flags the steps it cannot follow rather than silently reporting an impossible result. The other asserts that no sensitivity case is inert, which guards the trap described in Section 9.

## 12 Limitations

Four limitations bound the applicability of these results.

*No thermal model.* Continuous-power derating and the rise in battery internal resistance at low temperature are not represented. Neither binds on a legislative cycle, whose power demands are modest and brief, but both matter for sustained high-power driving and for cold starts.

*No battery ageing.* The results describe a new vehicle. Capacity fade and resistance growth over the life of the pack are outside the scope.

*Fitted rather than measured efficiency.* The drive-unit efficiency comes from the four-term loss model of Equation 7 fitted to plausible operating points, not from a measured map. The sensitivity study shows this contributes under 5% to the range prediction.

*Longitudinal dynamics only.* Lateral dynamics, suspension behaviour and the tyre slip that accompanies real cornering are not modelled, nor is charging behaviour or interaction with the grid.

## 13 Conclusions

A longitudinal model of the Tesla Model 3 RWD reproduces all three published performance figures from a single parameter set, with no figure tuned individually: acceleration to 100 km/h within 0.7%, top speed exactly, and WLTP range within 0.9%. Steady-cruise consumption, which entered no calibration, falls inside the measured real-world band at both 100 and 130 km/h.

Three findings are worth carrying forward.

**The vehicle is traction-limited off the line, not torque-limited.** The rear tyres, not the machine, set the acceleration up to 79 km/h, which is 77% of the time taken to reach 100 km/h. Dropping the tyre limit entirely is 12% optimistic; keeping it but omitting the dynamic load transfer is 12% pessimistic.

**Top speed is a software decision.** The electronic limiter binds well below both the machine’s mechanical speed ceiling and the point at which drag would overcome the available tractive force.

**The advertised drag coefficient does not predict range.** The coastdown-derived drag area exceeds the wind-tunnel figure by 29%, because a wind tunnel excludes wheel rotation and cooling airflow. Both numbers are correct, but

only one of them predicts energy, and using the textbook decomposition alone under-predicts road load by 12 to 18% above 80 km/h.

Finally, the sensitivity study shows that a result built on thirty-two estimated parameters can still be defensible, provided the estimates that dominate are identified. Here the auxiliary load dominates and the drivetrain assumptions do not, which means the prediction does not rest on the quantities that were hardest to determine.

## Acknowledgments

### Author Contributions

All authors contributed to the formulation of the vehicle model, the selection and justification of the assumed parameters, the interpretation of the sensitivity study, and the preparation of this manuscript. The software implementation, the validation campaign and the figure generation were carried out collaboratively. The authors thank the teaching staff of the *Electric Vehicle Technologies and Applications* module for guidance on vehicle longitudinal dynamics and legislative test procedures.

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### Conflicts of Interest

The authors declare no conflicts of interest regarding the publication of this article.

### Data Availability

All source code, the vehicle parameter file with the provenance of every value, the generated assumption register, and the scripts that regenerate every figure and table in this report are available in the open-source repository: <https://github.com/tarikbilla/Electric-Vehicle-Technologies-and-Applications>.

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